

Selective Electrosynthesis of Methanol from CO₂ Over Cu/Cu₂P₂O₇ Via the Formate Pathway

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The electrochemical CO₂ reduction reaction (CO₂RR) to methanol offers an eco-friendly approach to reducing carbon emissions while producing versatile liquid fuels and feedstocks. However, achieving high selectivity for methanol, especially at high current densities, remains challenging due to competing reactions that favor methane and hydrogen formation. Here, the tailored synthesis of Cu/Cu₂P₂O₇-based hybrid catalysts is reported for efficient and selective methanol production through the discharge of lithium-ion batteries. The catalyst exhibits a Faradaic efficiency exceeding 50% in both H-cells and gas-diffusion electrode cells, achieving one of the highest reported methanol partial current densities of over 100 mA cm⁻². Experimental and computational analyses reveal a synergistic effect between Cu nanoparticles with a predominant (111) surface and Cu₂P₂O₇ nanoparticles, which enhances selective methanol production via the HCOOH intermediate pathway. These findings provide insights into designing cost-effective electrocatalysts for selective methanol production.

sustainably producing valuable chemicals,^[2] including methane,^[3] ethylene,^[4] methanol,^[5] and ethanol.^[6] Among these products, methanol has garnered particular attention due to its dual role as both a fuel and feedstock.^[7] Despite significant recent advances in designing efficient electrocatalysts with high selectivity for various hydrocarbons^[3,4] and oxygenates,^[5,6] developing catalysts with high selectivity, especially for methanol, remains challenging. This difficulty arises mainly due to competing reactions, such as CO₂RR to methane^[8] and hydrogen evolution reactions (HER).^[9] Methanol and methane productions via CO₂RR share key intermediates, such as *HCO and *HCHO,^[10] which can either be converted to methanol through successive hydrogenation^[11] or to methane through deoxygenation followed by hydrogenation.^[12] To address this, researchers have suggested that an optimal *H coverage on catalysts is crucial for selectively producing methanol.^[13] However, excessive *H

1. Introduction

Electrochemical CO₂ reduction reactions (CO₂RR) offer a promising route for mitigating carbon emissions^[1] while

followed by hydrogenation.^[12] To address this, researchers have suggested that an optimal *H coverage on catalysts is crucial for selectively producing methanol.^[13] However, excessive *H

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coverage can promote HER, lowering the Faradaic efficiency for methanol production. Recently, promising electrocatalysts for methanol production, such as Ag₂S-Cu₂O/Cu,^[14] Rh₁Cu₄,^[15] and Cu/Au/NCF,^[16] have been reported. However, these often rely on expensive noble metals like Rh and Au, which are particularly effective for hydrogenation catalysis.^[17] Moreover, most of these catalysts show high methanol selectivity in non-aqueous electrolytes, where limited proton availability and low ionic conductivity present substantial obstacles to practical applications.^[18]

In this context, phosphate-based materials emerge as promising catalyst candidates for selective methanol production. Phosphates offer advantageous properties for CO₂RR, including polyprotic nature,^[19] exceptional chemical stability,^[20] and strong interaction with alkali metal ions like potassium ions,^[21] which can reduce the activation energy barrier of CO₂RR while suppressing HER. Previous studies have reported phosphate-based catalysts for selective formic acid production.^[22] However, despite the higher demand and value of methanol compared to formic acid, cost-effective Cu and phosphate-based electrocatalysts for methanol production have yet to be reported (Tables S1 and S2, Supporting Information), although many Cu-based thermocatalysts^[7b,c] for methanol production have been reported recently. This gap likely arises from the increased complexity of methanol production, which involves a six-electron, six-proton transfer process and requires optimal *H coverage to promote successive hydrogenation while suppressing HER. Therefore, precise control over the structure, composition, and *H coverage on phosphate-based catalysts may be essential for designing selective methanol catalysts.

In this study, we report the tailored electrochemical synthesis of Cu- and phosphate-based hybrid catalysts for selective methanol production via CO₂RR. Cu pyrophosphate (Cu₂P₂O₇) can be transformed into a mixture of Cu and Cu₂P₂O₇ nanoparticles with precisely tuned compositions through the discharge process of Li-ion batteries. Hybrid catalysts with the optimal composition exhibit high methanol selectivity and stability, achieving Faradaic efficiencies of 48.8% in 0.1 M KHCO₃ and 70.1% in 0.1 M CsHCO₃ in H-cells, while maintaining stable operation for 48 h. The hybrid catalyst also sustained its activity even on a gas-diffusion electrode (GDE), achieving the highest reported methanol partial current densities for Cu-based catalysts without noble metals, exceeding 100 mA cm⁻². In situ X-ray absorption spectroscopy and density functional theory (DFT) calculations suggest that the coexistence of Cu nanoparticles with (111) surface and Cu₂P₂O₇ nanoparticles enables selective methanol production through the HCOOH intermediate pathway. This study provides valuable insights into the design and optimization of cost-effective electrocatalysts for efficient CO₂RR, especially for selective methanol production.

2. Results and Discussion

2.1. Preparation and Characterization of the Hybrid Catalysts

We synthesized nanoparticulate hybrid electrocatalysts composed of metallic Cu and Cu₂P₂O₇ (CP) for selective methanol production, using pristine CP as a precursor. Unlike noble metals^[14–16] and Co,^[23] which are conventionally used in methanol production catalysts, Cu is both abundant and one of

the most active elements for CO₂RR. Additionally, phosphate offers beneficial properties for CO₂RR, such as its polyprotic nature and strong interaction with alkali metal ions. Based on these properties, we hypothesized that nanocomposites of Cu and phosphates could be designed as effective CO₂RR catalysts. CP, a promising next-generation cathode candidate in Li-ion batteries, was chosen as the catalyst precursor due to its ability to transform into a mixture of metallic Cu and Li₄P₂O₇ during the discharge process^[24] ($\text{Cu}_2\text{P}_2\text{O}_7 + 4\text{Li}^+ + 4\text{e}^- \rightarrow 2\text{Cu}^0 + \text{Li}_4\text{P}_2\text{O}_7$) (Figure 1a; Figure S1, Supporting Information). To enhance electrical conductivity, carbon nanotubes (CNTs) were incorporated into the catalyst design. As Li₄P₂O₇ is soluble in water, Cu/CP-based hybrid materials with tailored compositions were synthesized by discharging CP at a specified voltage, followed by washing with water (Figures S2 and S3, Supporting Information). The hybrid catalyst discharged at X.X V is denoted as CP-X.X. Electron microscopy confirmed the successful formation of Cu/CP nanocomposites (Figure 1b; Figures S3 and S4, Supporting Information). CP particles, initially 6.73 ± 1.91 nm in diameter, transformed into smaller Cu/CP nanocomposites with a size of 4.09 ± 1.23 nm during the discharge process, as CP partially transformed into Cu and soluble Li₄P₂O₇, with the latter removed by water (Figure S5, Supporting Information). Elemental mapping analysis, X-ray photoelectron spectroscopy (XPS), Auger spectroscopy, and infrared spectroscopy further verified the presence of Cu, P, and O elements in the catalysts, as well as their respective oxidation states (Figure 1c; Figure S6, Supporting Information). Notably, distinct Cu-rich nanoparticles were observed, with uniformly distributed O and P elements. These findings consistently demonstrate the successful formation of Cu/CP nanocomposites through the Li-ion battery discharge process.

2.2. Methanol Production by Electrochemical CO₂RR in H-Cells

We conducted electrochemical CO₂RR experiments using Cu foils coated with pristine CP or discharged CP. The CO₂RR experiments were performed in H-cells with 0.1 M KHCO₃ or 0.1 M CsHCO₃ electrolytes for 1 h (Figure 1d). For all pristine and discharged CP samples, the current density increased as the applied potential became more negative. In general, discharged CP samples exhibited higher current densities than pristine CP. However, when analyzing the current density profile at -1.2 V versus RHE, we found that the current density gradually decreased as the CP sample was discharged from CP-1.5 to CP-0.8 (Figures S7–S11 and Table S3, Supporting Information). Notably, all tested samples produced methanol, with CP discharged at lower voltages showing higher Faradaic efficiencies for methanol. While pristine CP achieved a Faradaic efficiency of 23.4% in 0.1 M KHCO₃ at -1.2 V versus reversible hydrogen electrode (RHE), CP-0.8 reached 48.8% under the same conditions. Using 0.1 M CsHCO₃, known to enhance CO₂ adsorption and improve CO₂RR selectivity^[25] further increased the Faradaic efficiency for methanol to 70.1% at -1.1 V versus RHE (Figure 1d; Figure S12 and Table S4, Supporting Information). Interestingly, CP-0.8 also produces a small amount of formic acid, which decreased as the cathodic potential and methanol production increased, suggesting methanol production via the formic acid pathway (Figure S13, Supporting Information). Due to its high selectivity,

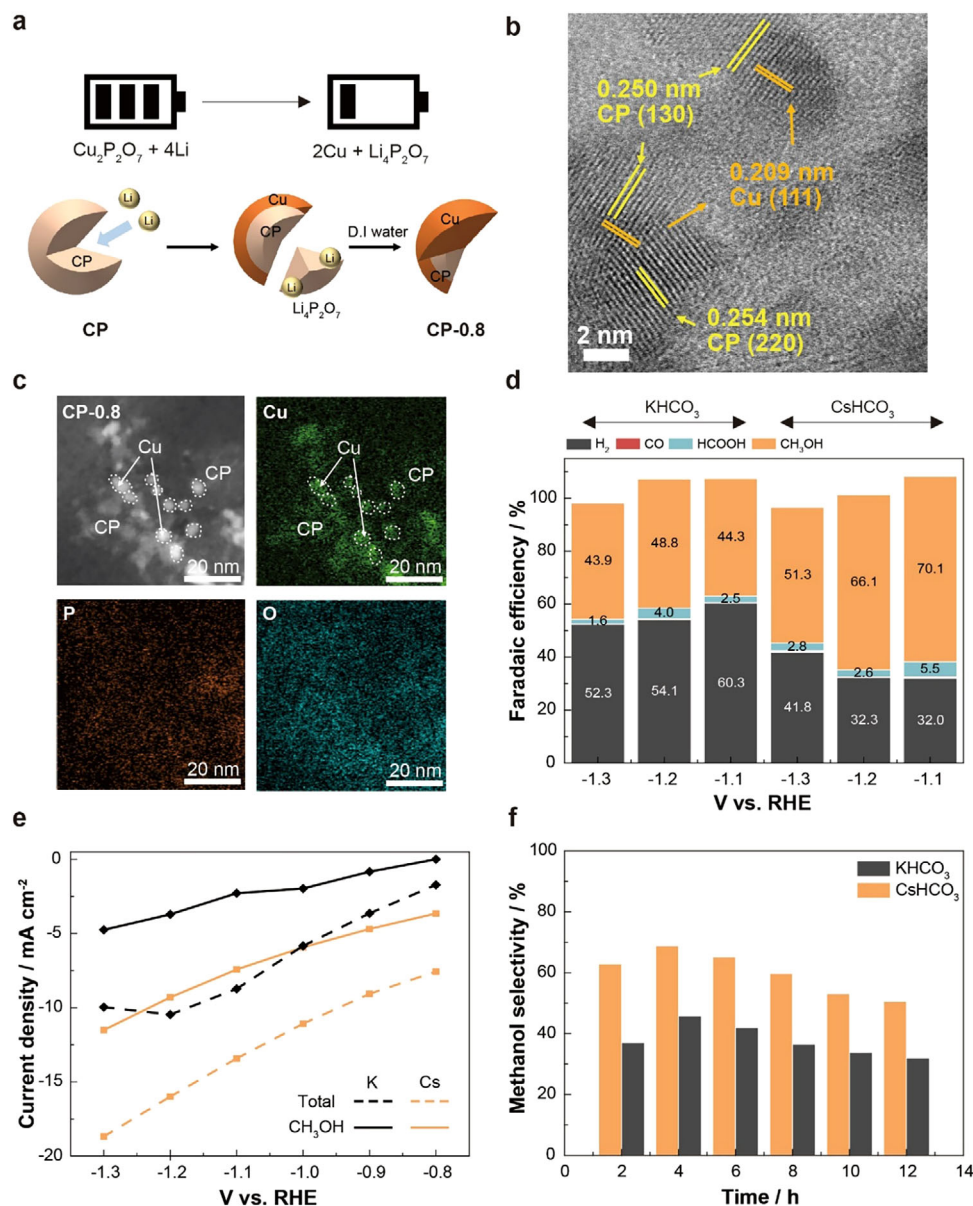


Figure 1. Synthesis and electrochemical CO₂RR performance of Cu/CP hybrid catalysts. a) Schematic illustration of the preparation of Cu/CP hybrid catalysts via the battery discharge process, followed by washing with water. b,c) Transmission electron microscopy (TEM) (b) and elemental mapping images (c) of CP-0.8. d,e) CO₂RR performance of the hybrid catalyst: (d) Faradaic efficiency and (e) methanol partial current density of CP-0.8 in 0.1 M KHCO₃ or 0.1 M CsHCO₃ electrolytes. f) Faradaic efficiency profile of CP-0.8 for methanol production at −1.2 V versus RHE.

CP-0.8 demonstrated an exceptionally high methanol partial current density, exceeding −5 to −10 mA cm^{−2} in H-cells (Figure 1e; Figure S14, Supporting Information). Additionally, CP-0.8 maintained a Faradaic efficiency greater than 50.0% at −1.2 V versus RHE in 0.1 M CsHCO₃ for >12 h (Figure 1f; Figure S15 and Tables S5 and S6, Supporting Information). It is worth noting that the observed gradual decrease in current density may be attributed to the depletion of dissolved CO₂ in the H-cells. Indeed, refreshing the electrolyte and intermittently purging it with CO₂ enabled sustained methanol production, confirming the robust catalytic performance of CP-0.8 (Figure S16 and Table S7, Supporting Information).

2.3. Active Site Identification by Ex Situ and In Situ Analyses

To identify the active sites of CP-0.8 for selective methanol production, we conducted various ex situ analyses, including XPS and X-ray diffraction (XRD). XPS analysis revealed multiple Cu oxidation states in both pristine and discharged CP samples: Cu 2p_{1/2} at 951–953 eV and Cu 2p_{3/2} at 932–934 eV, corresponding to Cu⁺ and Cu⁰ as well as Cu satellite peaks at 940–945 and 960–965 eV, indicative of Cu²⁺ (Figure 2a). With further discharging to 0.8 V, more Cu⁺ was reduced to metallic Cu, accompanied by the disappearance of Cu²⁺ and a decrease in P content (Figure S6b and Table S8, Supporting Information). XRD analysis showed a

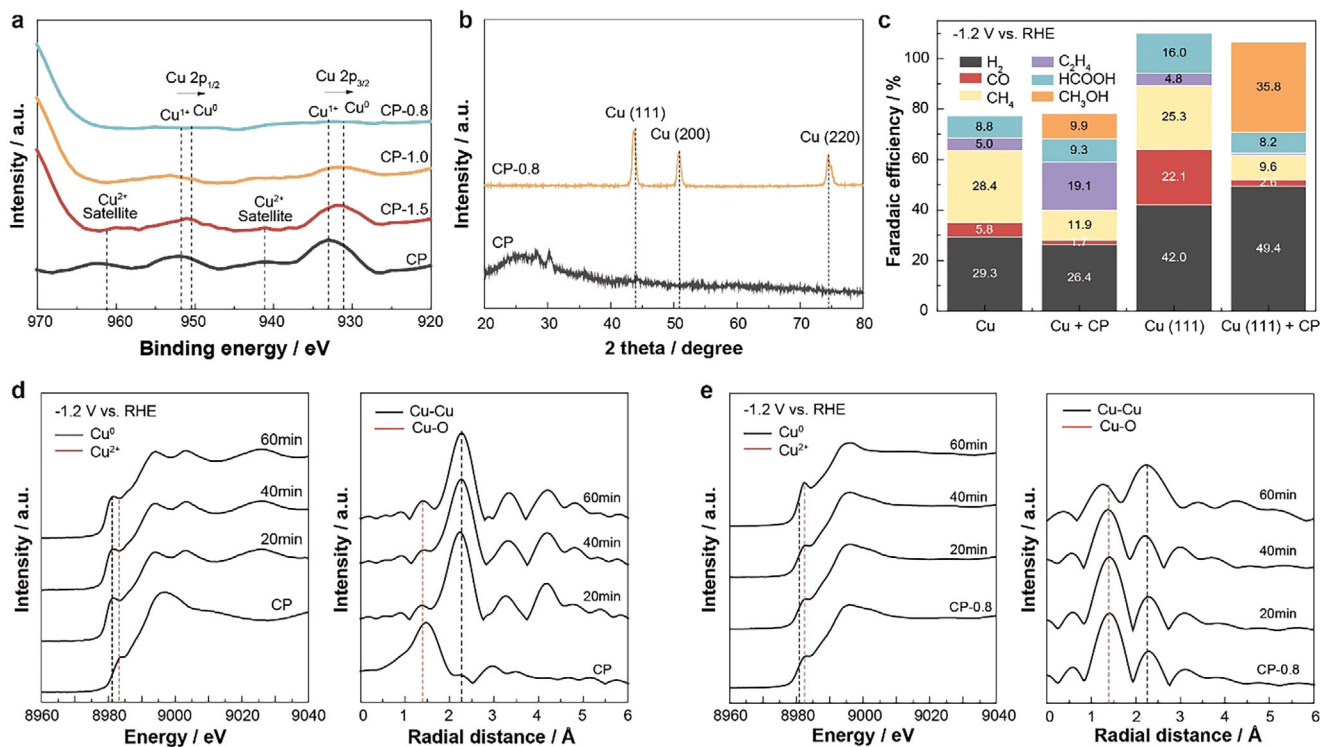


Figure 2. Ex situ and in situ analyses of the catalyst. a) XPS spectra of pristine CP and discharged CP catalysts. b) XRD patterns of pristine CP and CP-0.8. c) Faradaic efficiencies of polycrystalline Cu and single-crystalline Cu (111) with and without pristine CP modification. d,e) In situ X-ray absorption fine structure spectra of pristine CP (d) and CP-0.8 (e) in 0.1 M KHCO₃ at -1.2 V versus RHE.

prominent Cu (111) peak in CP-0.8, whereas pristine CP showed no distinct peaks (Figure 2b). Based on these findings, we hypothesized that the coexistence of Cu²⁺ and/or Cu¹⁺ in CP and metallic Cu⁰ in Cu (111) creates a synergistic effect, enhancing methanol selectivity.

To test this hypothesis, we conducted CO₂RR at -1.2 V versus RHE using polycrystalline Cu and single-crystalline Cu (111) foils prepared via contact-free annealing,^[26] both with and without pristine CP (Figure 2c; Figure S17 and Table S9, Supporting Information). Polycrystalline Cu or single-crystalline Cu (111) alone primarily produced methane, along with carbon monoxide, formic acid, and ethylene, as previously reported^[27] but no methanol. Interestingly, single-crystalline Cu (111) foil modified with CP produced methanol with a Faradaic efficiency of 35.8%, albeit lower than the 48.8% achieved by CP-0.8, which contains nanoparticulate Cu (111) and CP. Modifying polycrystalline Cu with CP also produced methanol, but at a lower selectivity, likely due to the presence of Cu (111) planes. These results suggest a synergistic effect between Cu (111) and CP at the nanoscale, enabling highly selective methanol production.

Given the dynamic changes often observed in Cu-based catalysts during CO₂RR^[28] and their associated stability challenges,^[29] we characterized pristine CP and CP-0.8 via in situ X-ray absorption spectroscopy to confirm active sites and stability. For pristine CP, oxidized Cu states were rapidly reduced to metallic Cu within the first 20 min, accompanied by bond length changes reflecting the conversion of most Cu–O bonds to Cu–Cu bonds (Figure 2d). Interestingly, a Cu (111) peak also

appeared in pristine CP after CO₂RR, potentially explaining the small amount of methanol produced by pristine CP (Figure S18, Supporting Information). In contrast, CP-0.8 maintained its Cu oxidation state and Cu–O bonds throughout the CO₂RR, after an initial slight reduction of oxidized Cu to metallic Cu (Figure 2e). These findings underscore the significance of tailoring the Cu and CP composition for enhanced stability and suggest that the optimal CP content in CP-0.8 supports improved stability, as discussed further in a subsequent section. Notably, this enhanced stability is also attributed to the stronger binding of P₂O₇⁴⁻ to transition metals compared to O²⁻. (e.g., in CuO and Cu₂O, which are commonly used in CO₂RR but suffer from poor stability).^[30]

2.4. Experimental and Theoretical Verification of the Reaction Pathway for Methanol Production

To elucidate the reaction pathway for efficient methanol production, we conducted electrochemical reactions using various feedstocks, including CO₂, CO, and HCOOH. According to the literature,^[31] formaldehyde (*HCHO) is a key intermediate in methanol production via CO₂RR and can be formed through two distinct pathways: the *CO pathway and the *HCOOH pathway (Figure 3a). For CP-0.8 catalysts, significant amounts of methanol were produced when CO₂ and formic acid were used as feedstocks, whereas only negligible methanol was generated when CO was used as the feedstock

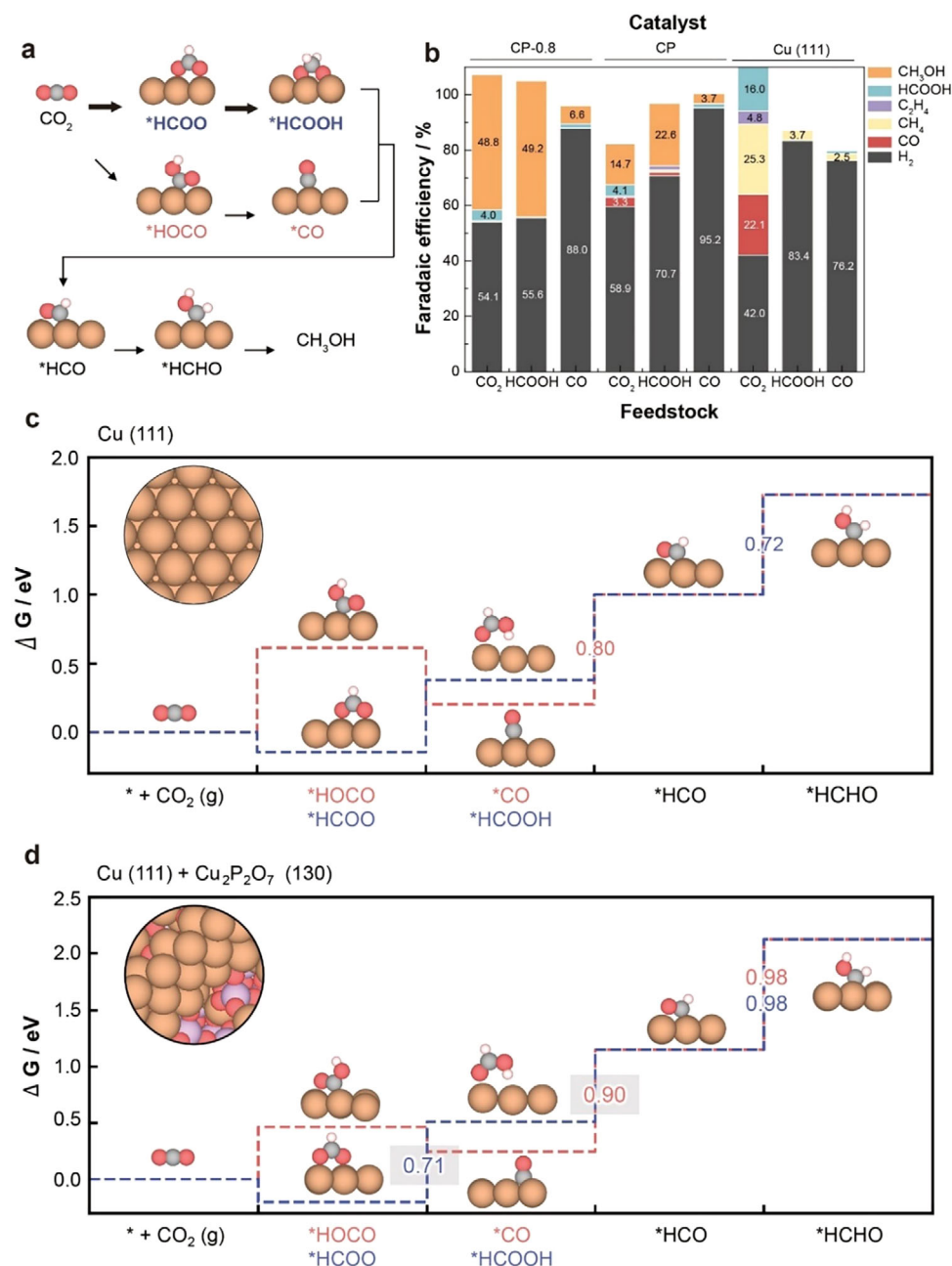


Figure 3. Experimental and computational investigation of the reaction pathways. a) Schematic illustration of pathways to form *HCHO intermediates crucial for methanol production via CO₂RR. b) Faradaic efficiency of CP-0.8 at -1.2 V versus RHE in a 0.1 M KHCO₃ electrolyte with different feedstocks: CO₂, HCOOH, and CO. c, d) DFT-calculated Gibbs energy diagrams for the CO pathway (red line) and HCOOH pathway (blue line) on Cu (111) (c) and Cu (111) + Cu₂P₂O₇ (130) (d) surfaces. The Gibbs energy values of the rate-determining step are reported within the figure, with the second highest energy values shaded in grey shadow. Cu, O, C, H, and P atoms are represented by orange, red, grey, white, and purplish-grey, respectively.

(Figure 3b; Figure S19 and Table S10, Supporting Information). Interestingly, methanol production from CO₂ and formic acid was also observed when using pristine CP as a catalyst. In contrast, Cu (111) alone produced no methanol, regardless of the feedstock type, although a small amount of formic acid was generated when using CO₂. These findings suggest a synergistic effect between Cu (111) and CP in CP-0.8, with Cu (111) responsible for formic acid formation as an intermediate and CP

promoting its subsequent conversion to methanol. This aligns with our observation that discharged CP produces HCOOH but not CO at low applied potentials (Figure S13, Supporting Information). This result is noteworthy, as conventional studies have predominantly emphasized the *CO pathway for methanol production.^[32]

To understand the structure of discharged CP and the catalytic mechanism at the atomic scale, we performed DFT calculations

to elucidate their surface structures and associated physicochemical properties. For modeling the Cu (111) + Cu₂P₂O₇ (130) interface, consistent with high-resolution TEM and fast Fourier transform analysis (Figure 1b; Figures S4 and S5, Supporting Information), we used the *Cellmatch* code^[33] to align Cu₂P₂O₇ (130) with Cu (100), achieving a strain rate of less than 2%. We then constructed the Cu (111) + Cu₂P₂O₇ (130) interface by introducing additional Cu atoms (Figure S20a, Supporting Information). After structure optimization, we found that the bottom layers of the Cu (111) surface were mildly reconstructed/oxidized (Figure S20b,c, Supporting Information). Radial distribution function analysis revealed that the main peak of the reconstructed Cu (111) surface was broadened compared to the ideal bulk-truncated surface, and the first peak position shifted slightly lower (Figure S21, Supporting Information). We did not consider Cu₂P₂O₇ (130) as an active site, because it exhibited significantly higher Gibbs energy (0.42–1.15 eV) for CO adsorption, compared to Cu (111) (−0.41 eV), and in some instances, CO was detached from the surface upon relaxation (Figure S22, Supporting Information).

Next, we investigated the catalytic mechanism on both clean Cu (111) (Figure 3c) and Cu (111) + Cu₂P₂O₇ (130) slabs (Figure 3d) to better understand the role of Cu₂P₂O₇ in the catalytic reaction. According to the literature^[8b,34] the CH₂O₂ adsorbs in several geometrical configurations, specifically as H₂COO, HCOOH-1, and HCOOH-2. Using these configurations, we found that their total energy differences fall within 6–8 meV atom^{−1}. Since experimentally identifying the preferred adsorption geometry is challenging, our computational model included all three binding geometries, and we report the configuration with the lowest Gibbs free energy for the potential-determining step (PDS). Results for other geometries are presented in Figure S23 (Supporting Information).

On a clean Cu (111) surface, the rate-determining step for the CO pathway was *CO → *HCO, with a Gibbs energy difference of 0.80 eV. For the CH₂O₂ pathway, the rate-determining step was either *HCOO → *HCOOH or *HCO → *HCHO, with Gibbs energy differences ranging from 0.72 to 0.94 eV (Figure 3c; Figure S23a, Supporting Information). The Gibbs energy differences for the PDS in the HCOOH pathway were within 0.1 eV of the CO pathway, while the H₂COO pathway exhibited a higher energy difference of ≈0.14 eV compared to the CO pathway. This suggests that the catalytic reactions involving CO are energetically competitive on clean Cu (111), aligning with experimental results (Figure 2c).

In contrast, on the Cu (111) + Cu₂P₂O₇ (130) slab, the rate-determining steps for the CO and CH₂O₂ pathways are the same, namely *HCO → *HCHO, except for the H₂COO path. The H₂COO pathway was excluded due to its high Gibbs free energy barrier (1.92 eV), indicating that the reaction is unlikely to proceed via this route (Figure S23b, Supporting Information). Since the CO and other CH₂O₂ pathways share the same rate-determining steps, we investigated the second-highest energy steps. For the CO pathway, the second-highest energy step is *CO → *HCO, with a Gibbs energy of 0.90 eV. In the case of the HCOOH-1 pathway, the Gibbs energy (0.83 eV) is comparable to that of the CO pathway; however, the HCOOH-2 pathway shows a significantly lower Gibbs energy, with a difference of ≈0.2 eV (Figure 3d). In the Cu (111) + Cu₂P₂O₇ (130) configuration, the CO pathway now requires a higher energy barrier, which explains

why the Faradaic efficiency for methanol under a CO atmosphere is noticeably low (Figure 3b).

For a simplified model, we also investigated Cu (111) + PO₄. First, we assessed the relative thermodynamic stability as a function of the number of PO₄ adsorptions and pH conditions (Figures S24–S26, Supporting Information). We created surface slab models with 1–4 PO₄ molecules on (4 × 4) Cu (111) slabs and calculated the Gibbs free energy of adsorption (ΔG₂) under varying pH conditions, following previous studies.^[35] Further details on the calculations are provided in the Supplementary Information. Our calculation revealed that the Cu (111) surface with a small amount of PO₄ is the most thermodynamically stable. This finding supports the superior long-term stability of CP-0.8 (Figure 2e), which contains only small amounts of PO₄. On Cu (111) + PO₄, there is a clear preference for the CH₂O₂ pathway over the CO pathway. All CH₂O₂-related pathways exhibited lower Gibbs energies for the PDS compared to the CO pathway, with absolute differences exceeding 0.1 eV (Figure S22c, Supporting Information). Notably, in the H₂COO and HCOOH-1 cases, the energy differences were greater than 0.2 eV. These findings further support the critical roles that both the Cu (111) surface and phosphate-related species play in altering the reaction pathway.

To further confirm the proposed reaction pathway, *operando* Raman spectroscopy was conducted to identify key intermediates (Figure S27, Supporting Information). However, distinguishing the characteristic Raman peak for formate species (≈1300–1400 cm^{−1}) was challenging due to interference from the D-band of carbon incorporated during the catalyst synthesis (≈1350 cm^{−1}).^[36] Nevertheless, in situ IR spectroscopy clearly identified the adsorbed formate (*HCOO) species, showing distinct characteristic peaks ≈1390 and 1280 cm^{−1}, along with an additional shoulder peak at ≈1570 cm^{−1} (Figure S28, Supporting Information).^[37] The formate species rapidly decreased at higher potentials, indicating the consumption of formate species for facilitating methanol production. These IR results highly support the formation and adsorption of formate intermediate on the catalyst surface, emphasizing the validity of the proposed HCOOH reaction pathway.

2.5. Methanol Production by CO₂RR Under Various pH Conditions and in GDE Cells

To evaluate methanol selectivity under different pH conditions, we prepared electrolytes with pH values of 4.0, 7.0, 9.0, 11.0, and 13.0 (Figure 4a,b; Table S11, Supporting Information). While pristine CP predominantly produced hydrogen across these pH conditions, CP-0.8 consistently showed high methanol selectivity, with Faradaic efficiencies exceeding 30% (Figure S29, Supporting Information). In particular, CP exhibited the highest Faradaic efficiency for methanol near neutral pH, likely due to increased CO₂ availability (not present in the form of carbonate) and reduced competition from C₂ product pathways, as previously reported.^[38] The pH dependence of methanol production showed a similar trend regardless of the feedstock—CO₂, HCOOH, or CO (Figure S30, Supporting Information). These results highlight the superior catalytic activity and stability of CP-0.8 for selective methanol production in CO₂RR across a wide

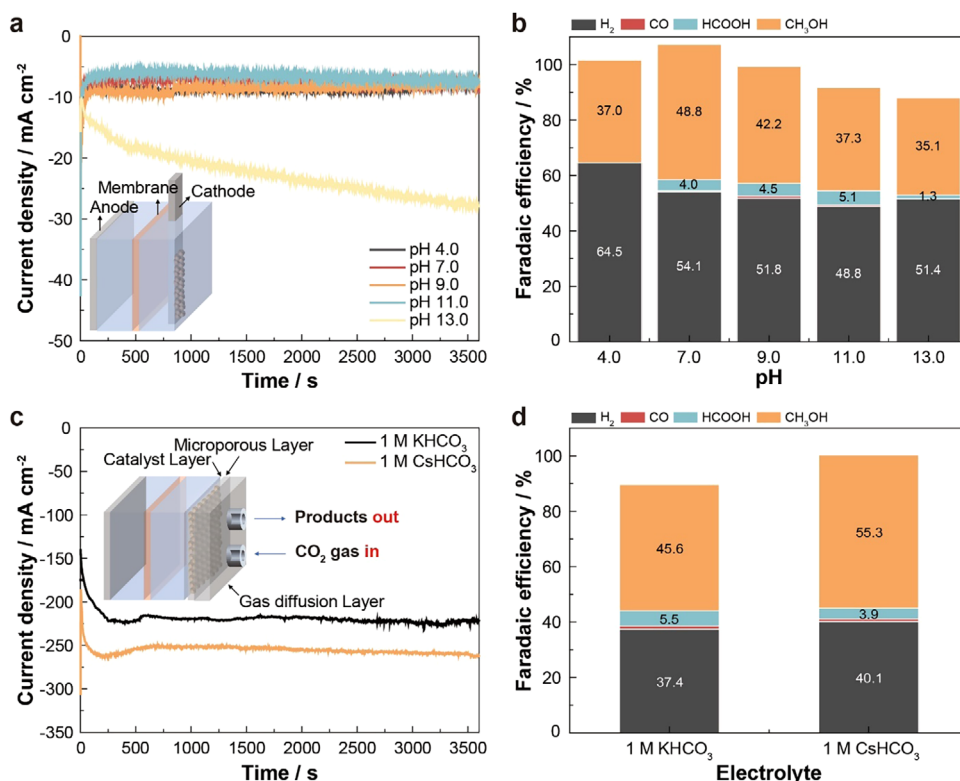


Figure 4. Electrochemical performance under various pH conditions and in a flow cell. a, b) Current density (a) and Faradaic efficiency (b) of CP-0.8 under different pH conditions (Inset: schematic of the H-cell setup). c, d) Current density (c) and Faradaic efficiency (d) of CP-0.8 at -1.0 V versus RHE in 1 M KHCO_3 and 1 M CsHCO_3 electrolytes (Inset: schematic of the GDE setup).

pH range, suggesting a consistent reaction pathway. Nevertheless, further studies are required to fully validate these findings. Notably, CP-0.8 displayed peak activity near neutral pH, close to its pKa value, suggesting that an optimal degree of protonation and/or adequate $^*\text{H}$ coverage are critical for efficient methanol production.

To further enhance the methanol production rate, we employed a GDE in a flow cell setup. At -1.0 V versus RHE, the GDE cells using CP-0.8 achieved methanol Faradaic efficiency of 45.6% in 1 M KHCO_3 and 65.7% in 1 M CsHCO_3 (Figure 4c,d; Figures S31 and S32 and Tables S12 and S13, Supporting Information). The partial current density for methanol production with CP-0.8 in GDEs reached 113.3 mA cm^{-2} in 1 M KHCO_3 and 145.5 mA cm^{-2} in 1 M CsHCO_3 , significantly surpassing the performance of pristine CP in GDE cells (23.9 mA cm^{-2}) and CP-0.8 in H-cells (Figure S31, Supporting Information). To the best of our knowledge, a methanol partial current density exceeding 100 mA cm^{-2} represents one of the highest values reported to date for catalysts without noble metals and the highest among Cu-based catalysts in this category. In particular, CP-0.8 showed the best Faradaic efficiency for methanol of 65.7% at -0.8 V versus RHE (Figure S33 and Tables S14–S16, Supporting Information).

The $\text{Cu/Cu}_2\text{P}_2\text{O}_7$ (Cu/CP)-based hybrid catalyst represents a significant breakthrough in CO_2 reduction reaction for methanol production. This catalyst not only achieves one of the highest methanol partial current densities reported for Cu-based elec-

trocatalysts, exceeding 100 mA cm^{-2} , but also does so with remarkable efficiency, making it stand out among non-noble metal-based catalysts (Tables S15 and S16, Supporting Information). Its cost-effectiveness and scalability further enhance its potential, as the Cu/CP catalyst operates efficiently in aqueous electrolytes with high Faradaic efficiency. The catalyst's performance in gas diffusion electrode (GDE) systems offers additional advantages, making it a promising candidate for large-scale industrial applications. This advancement holds significant implications for CO_2 RR research, as the Cu/CP catalyst offers a sustainable, high-performance solution for methanol synthesis. Methanol, a highly valuable renewable fuel and versatile chemical feedstock, plays a crucial role in producing industrial chemicals, including aldehydes, olefins, and resins. With its exceptional efficiency, cost-effectiveness, and compatibility with GDE systems, the Cu/CP catalyst is poised to contribute significantly to the industrial adoption of CO_2 reduction technologies, particularly in the fields of renewable fuels, chemical manufacturing, and energy storage.

Nevertheless, there are still some limitations that require further investigation for practical application. For example, despite significant progress, the selectivity and current density for methanol production remain lower than those achieved for other CO_2 RR products such as CO, formic acid, and C_2H_4 , which exhibit Faradaic efficiencies approaching or even exceeding 70–80% and partial current densities ranging from several hundred mA cm^{-2} to over 1 A cm^{-2} . Catalyst stability must also be evaluated and enhanced beyond the laboratory scale (Figure S16,

Supporting Information). Additionally, the current synthesis method relies on battery discharge cycles, which limits scalability. Therefore, alternative synthesis approaches based on the design principles proposed in this study should be further explored to enable large-scale application.

3. Conclusion

In summary, we synthesized Cu/CP-based hybrid catalysts for efficient and selective methanol production via CO₂RR. These hybrid catalysts were tailored through a discharge process of Li-ion batteries, with the CP-0.8 catalyst demonstrating the highest efficiency and stability for methanol production. Ex situ analysis revealed that the superior catalytic activity of CP-0.8 arises from a synergistic effect between metallic Cu nanoparticles with a predominant (111) surface and CP nanoparticles, which facilitate formic acid formation as an intermediate and its subsequent conversion to methanol, respectively. In situ X-ray absorption spectroscopy further confirmed the stable presence of both metallic and oxidized Cu species in CP-0.8 during CO₂RR. Electrochemical reactions using various feedstocks, supported by DFT calculations, demonstrated that methanol production proceeds via the HCOOH pathway on Cu (111) in the presence of phosphate. CP-0.8 catalysts exhibited a high Faradaic efficiency of 50–70% and one of the highest reported methanol partial current densities exceeding 100 mA cm⁻² in GDE cells. These findings provide valuable insights into the design and synthesis of novel CO₂RR electrocatalysts, offering a promising pathway for efficient and sustainable methanol production.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

Cu/Cu₂P₂O₇ catalyst, electrochemical CO₂ reduction reaction, HCOOH pathway, methanol production

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